Software Requirements Specification (SRS)

Drug Interaction and Safety Alert System

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1. Introduction

1.1 Purpose

This document defines the software requirements for the Drug Interaction and Safety Alert System.
The goal of this system is to detect potential drug interactions and generate alerts for patients or medical staff.

1.2 Scope

This project focuses on detecting dangerous drug combinations and alerting the user.
It includes modules for drug interaction checking, database management, user input validation, alert notification, and basic logging.

1.3 Definitions, Acronyms, and Abbreviations

- Drug Interaction: An effect resulting from the combination of two or more drugs.
- Alert: A warning generated to inform about the danger.
- DB: Database storing drug interaction data.
- UI: User Interface

1.4 References

- IEEE SRS Standards
- PlantUML, SPIN, Python, Pytest official documentation

1.5 Overview

The rest of this document describes the product, its features, constraints, interfaces, and requirements.

2. Overall Description

2.1 Product Perspective

This is a standalone system used by patients and healthcare staff for safety analysis of drug prescriptions.

2.2 Product Functions

- Input drug names by user.
- Check if interactions exist in the database.
- Generate alert if interaction is detected.
- Admin module to update interaction database.

2.3 User Classes and Characteristics

- Patients: Basic UI users with minimal technical background.
- Healthcare Staff: Users who check interactions for prescriptions.
- Admin: Manage drug interaction entries.

2.4 Operating Environment

Python 3.x, Windows/Linux, CLI/GUI frontend (optional), local files for DB

2.5 Design and Implementation Constraints

Only Python standard libraries and simple data formats (like dicts or JSON) are used.

2.6 User Documentation

Includes README with instructions, test cases, and usage examples.

2.7 Assumptions and Dependencies

Assumes correct spelling of drug names, pre-populated DB, internet not required.

3. System Features

3.1 Login

Allows authorized users (Doctors, Pharmacists, System Administrators) to securely access the system with authentication credentials. Each login attempt is logged for security and auditing.

3.2 Enter Prescription

Doctors input patient prescriptions, which are stored and cross-checked for drug interactions. Prescriptions include patient ID, drug name, dosage, and frequency.

3.3 Check Drug Interaction

This module checks the entered prescription against known drug interactions in the drug database and triggers alerts if risky combinations are found.

3.4 Alert Drug Interaction

If a potential harmful drug combination is found, the system generates an alert specifying the drugs involved, severity, and guidance.

3.5 Notify User

Notifies the doctor and patient via the user interface or configured notification service (e.g., email, SMS) about any drug interactions detected.

3.6 Receive Alerts

Allows users (Doctors, Patients) to view alerts on the dashboard in real-time. Alerts remain logged and accessible for reference.

3.7 Generate Reports

The system administrator or doctor can generate reports including patient records, prescriptions, interactions, and activity logs.

3.8 Manage User Accounts

System administrators manage user accounts, including creating, updating, disabling accounts, and assigning roles and permissions.

3.9 View Patient History

Doctors can view a patient's historical medical records, including past prescriptions, diagnoses, and interaction alerts.

3.10 Manage Patient Records

Doctors and authorized personnel can update patient details, medical history, and maintain electronic health records securely.

3.11 Update Drug Information

Pharmacists are responsible for updating the internal drug database with new drugs, dosage guidelines, or contraindications.

3.12 Access Drug Database

Grants permission to query the drug database for detailed drug profiles, interaction pairs, and safety warnings.

3.13 Sync Drug Data

Automatically synchronizes the internal drug data with the external drug database periodically to ensure up-to-date safety alerts.

3.14 Log Activity

Logs all critical actions such as login, prescription entries, data updates, and alerts for auditing and troubleshooting.

4. External Interface Requirements

4.1 User Interfaces

Command-line interface (CLI); GUI optional.

4.2 Hardware Interfaces

None specifically required; standard PC setup sufficient.

4.3 Software Interfaces

Uses standard Python libraries. May interface with JSON or flat file for DB.

4.4 Communication Interfaces

Not applicable (standalone system).

5. Other Non-Functional Requirements

5.1 Performance Requirements

System should detect and respond within 1 second for input of 1–5 drugs.

5.2 Safety Requirements

Alerts must not be missed. System should clearly show warnings to users.

5.3 Security Requirements

Only admins should have access to modify the database.

5.4 Software Quality Attributes

Maintainability, usability, and reliability are prioritized.